# THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

## CLASS VIII — SCIENCE · MID-TERM PRACTICE PAPER

Session 2026-27 · Term II · Chapters 8–13

| Class: 8 | Subject: Science | Set 04 | Time: 2 Hours 30 Minutes | Max. Marks: 60 |
|---|---|---|---|---|

**General Instructions:**

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

### SECTION A — Multiple Choice Questions — 1 questions · 12 marks

*Question 1 has twelve parts (I–XII). Each carries 1 mark.*

**1. From Question I–XII, choose the correct option.**

I. Which of the following is an element? 1

(a) Sulphur (b) Common salt (c) Water (d) Sugar

II. Which of these is a property of most non-metals? 1

(a) Sonorous (b) Poor conductors of heat and electricity

(c) Lustrous and ductile (d) Malleable

III. Which of these is a mixture? 1

(a) Carbon dioxide (b) Oxygen (c) Pure water (d) Air

IV. Iron filings can be separated from a mixture of iron filings and sand by: 1

(a) evaporation (b) sublimation

(c) magnetic separation (d) filtration

V. A mixture of water and kerosene oil can be best separated using: 1

(a) a separating funnel (b) a magnet

(c) sieving (d) sublimation

VI. In a salt solution, salt is the: 1

(a) solute (b) solution (c) solvent (d) precipitate

VII. A solution in which no more solute can dissolve at a given temperature is called: 1

(a) saturated solution (b) unsaturated solution

(c) suspension (d) dilute solution

VIII. The solubility of most solids in water generally: 1

(a) becomes zero on heating (b) decreases with temperature

(c) increases with increase in temperature (d) does not depend on temperature

IX. The image formed by a plane mirror is: 1

(a) real and inverted

(b) virtual, upright and of the same size as the object

(c) virtual and smaller

(d) real and magnified

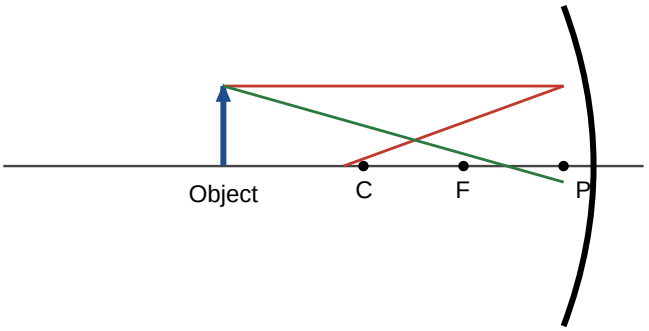
X A ray of light strikes a plane mirror at an angle of incidence of 35°. The angle between the incident ray and the reflected ray is: 1

(a) 35° (b) 110° (c) 70° (d) 55°

XI A shaving mirror gives an enlarged, upright image. It is a: 1

(a) plane mirror (b) concave mirror (c) convex mirror (d) concave lens

XII In the figure the object is placed beyond C of a concave mirror. The image is formed: 1




(a) at the pole, virtual (b) at infinity

(c) behind the mirror (d) between F and C, real and inverted

## SECTION B — Very Short Answer Questions — 4 questions · 8 marks

*2 marks each (about 30–40 words).*

**2.** Name two metals and two non-metals. **2**

**OR**

Why are aluminium and copper used for making cooking vessels and electric wires respectively?

**3.** Differentiate between an element and a compound with one example each. **2**

**4.** What is meant by solubility? Name one factor that affects it. **2**

**5.** State the two laws of reflection of light. **2**

## SECTION C — Short Answer Questions — 3 questions · 9 marks

*3 marks each (about 60 words).*

**6.** Describe how you would separate a mixture of salt and sand. Name the techniques used. **3**

**OR**

How can ammonium chloride be separated from a mixture of ammonium chloride and sand? Name the process and give the reason.

**7.** Rohan added 10 g of sugar to 50 mL of water and stirred. Some sugar remained undissolved. What does this show? What can he do to dissolve it? Name the solute and the solvent. **3**

**8.** Distinguish between a concave mirror and a convex mirror on the basis of reflecting surface and one use of each. **3**

**OR**

A convex lens is used as a magnifying glass. Where should the object be placed to get an enlarged image? Why is a convex lens called a converging lens?

## SECTION D — Long Answer Questions — 3 questions · 15 marks

*5 marks each.*

**9.** Complete the given sentences (choose from the words in the box): **(Sublimation, Saturated, Dispersion, Lustre, Alloy)** **5**

i. The shine of the surface of a metal is called _____.
ii. A mixture of two or more metals, or a metal and non-metal, is an _____.
iii. The change of a solid directly to a gas on heating is _____.
iv. A _____ solution cannot dissolve any more solute at that temperature.
v. The splitting of white light into its colours is _____.

**10.** Match the Columns **5**

| Column I | Column II |
|---|---|
| (a) Filtration | (i) separating the components of ink or dyes |
| (b) Distillation | (ii) separating insoluble solid from a liquid |
| (c) Chromatography | (iii) separating large from small particles |
| (d) Sieving | (iv) obtaining pure water from salt water |
| (e) Winnowing | (v) separating light husk from heavier grains |
| | (vi) separating a magnetic substance |

**11.** (a) Differentiate between a real image and a virtual image (any three points). (3) **5**
(b) Why does a concave mirror used by a dentist show an enlarged image? (2)

**OR**

(a) With a labelled sketch, show reflection of a ray at a plane mirror. (3)
(b) What is the angle of reflection if the incident ray makes 30° with the mirror? (2)

## SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

*Read the cases carefully and answer. Each question carries 4 marks.*

**12.** Meena stirred some soil in a beaker of water. After standing for a while, the soil settled at the bottom and the water on top was clear. She then poured the water through filter paper.

(a) What is the mixture of soil and water called? 1

(b) Name the method used for separating clear water from the settled soil by pouring. 1

(c) Meena wants to obtain pure water from salty sea water. Which method will she use? Give the principle. 2

**OR**

(d) Why is the filtrate of the soil-water mixture still not completely pure?

**13.** A student measured how much of a certain salt dissolved in 100 mL water at different temperatures: 20 °C – 36 g; 40 °C – 37 g; 60 °C – 38 g; 80 °C – 40 g.

(a) What happens to the solubility of salt as temperature rises? 1

(b) How much salt will dissolve in 100 mL of water at 60 °C? 1

(c) A solution at 20 °C has 30 g salt in 100 mL water. Is it saturated, unsaturated or supersaturated? Why? 2

**OR**

(d) Suggest one way to dissolve more sugar in a given quantity of water.

**14.** Ria placed a pencil in front of three mirrors – plane, concave and convex – and observed the images. In the plane mirror the image was the same size. In the second mirror the image was larger. In the third the image was smaller.

(a) Which mirror produced the larger image? 1

(b) Which mirror produced a smaller, upright image? 1

(c) Name one use of a convex mirror and give a reason. 2

**OR**

(d) Why does a person see lateral inversion of the letters in a plane mirror?

**15.** A teacher shows two samples: a shining yellowish metal rod that can be beaten into a sheet, and a dull brittle yellow solid that breaks when hammered.

(a) What property of the first sample allows it to be beaten into sheets? 1

(b) Identify the second sample as a metal or a non-metal. 1

(c) Give two more differences between metals and non-metals. 2

**OR**

(d) Name a metal that is liquid at room temperature and a non-metal that is a good conductor of electricity.